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A Comprehensive Survey of Solar Power Forecasting Methods: Machine Learning, Deep Learning, and Hybrid Approaches (2020–2025)

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Abstract

Solar photovoltaic (PV) power forecasting has emerged as a critical enabler of renewable energy grid integration, attracting significant research attention over the past five years. This paper presents a comprehensive survey of solar power forecasting methods published between 2020 and 2025, systematically reviewing more than 20 peer-reviewed studies spanning statistical models, classical machine learning (ML), deep learning (DL), and hybrid ensemble approaches. For each category, we analyze the modeling methodology, dataset characteristics, evaluation metrics, reported accuracy, and key limitations. A unified comparison table summarizes performance across studies in terms of Root Mean Square Error (RMSE) and coefficient of determination (R^2). From this systematic review, we identify five persistent research gaps: the widespread neglect of night-time zero-production records during training, the absence of structured feature engineering beyond raw meteorological inputs, the under-treatment of right-skewed target distributions, the limited focus on multi-horizon accuracy beyond 24 hours, and the lack of ablation studies that isolate the contribution of individual modeling decisions. These gaps motivate future work combining pre-modeling optimization with state-of-the-art gradient boosting and hybrid architectures. This survey is intended to serve as a reference for researchers entering the solar forecasting domain and as a foundation for positioning novel contributions against the existing body of work.

Index Terms— Solar power forecasting, photovoltaic systems, machine learning, deep learning, LSTM, random forest, gradient boosting, ensemble methods, literature review, renewable energy.

1 Introduction

The rapid expansion of solar photovoltaic (PV) generation has transformed the global energy landscape. Global installed PV capacity surpassed 1 terawatt (TW) in 2022 and is projected to reach 3 TW by 2030 [1], driven by declining module costs and supportive energy policies. However, the inherent variability and weather dependence of solar generation pose significant challenges for power grid operators, energy traders, and plant managers who must balance supply and demand in real time.

Accurate solar power forecasting across multiple time horizons—from intra-hour nowcasting to 10-day-ahead planning—is therefore a fundamental requirement for reliable and economical solar energy integration. One-day-ahead forecasts support day-ahead electricity market bidding and spinning reserve scheduling. Three-to-five-day forecasts enable unit commitment planning and preventive maintenance scheduling. Ten-day forecasts inform seasonal energy storage dispatch and demand-side management strategies [2].

The challenge of solar forecasting lies in the complex, non-linear relationships between meteorological variables—solar irradiance, temperature, humidity, wind speed, and cloud cover—and photovoltaic power output. Traditional statistical methods such as ARIMA and persistence models are inadequate for capturing these non-linear dynamics [3]. This limitation has driven a wave of research applying ML and DL techniques to the problem, producing a rich but fragmented literature.

This survey makes the following contributions:

1. A systematic review of 20 solar power forecasting studies published between 2020 and 2025, organized by modeling paradigm.
2. A unified performance comparison table enabling cross-study benchmarking.
3. Identification of five persistent research gaps that the existing literature has not adequately addressed.
4. A forward-looking discussion of promising research directions.

The remainder of this paper is structured as follows. Section 2 defines the survey scope and methodology. Section 3 reviews statistical and persistence models. Section 4 surveys classical machine learning approaches. Section 5 covers deep learning methods. Section 6 discusses ensemble and hybrid approaches. Section 7 provides a unified cross-study comparison. Section 8 identifies research gaps. Section 9 outlines future research directions. Section 10 concludes the survey.

2 Survey Scope and Methodology

2.1 Inclusion Criteria

Papers were selected from IEEE Xplore, ScienceDirect, Web of Science, and Google Scholar using the search terms: "solar power forecasting," "photovoltaic forecasting," "solar irradiance prediction," "machine learning solar energy," and "deep learning PV generation." The review covers publications from January 2020 to December 2025. Inclusion required: (i) peer-reviewed publication in a journal or conference proceedings; (ii) quantitative evaluation using RMSE, MAE, or R^2; and (iii) application to solar PV power or irradiance forecasting. Review papers, position papers, and studies without quantitative benchmarks were excluded.

2.2 Classification Framework

Reviewed papers are classified into four categories: (1) statistical and persistence models, (2) classical machine learning (SVM, RF, gradient boosting), (3) deep learning (LSTM, CNN, attention, transformers), and (4) ensemble and hybrid approaches combining multiple modeling paradigms. Within each category, papers are discussed in chronological order to highlight trends over the review period.

2.3 Performance Metrics

The primary metrics used for cross-study comparison are:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{1}$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{2}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{3}$$

where y_i are actual values, $\hat{y}_i$ predicted values, and $\bar{y}$ the mean of actual values. MAPE is also reported where available.

3 Statistical and Persistence Models

3.1 Overview

Statistical and persistence models represent the earliest class of solar forecasting approaches and continue to serve as baselines against which more sophisticated methods are evaluated. The persistence model assumes that future irradiance or power output equals current values, offering a trivially simple but surprisingly competitive benchmark for very short-horizon (sub-hourly) forecasts.

3.2 ARIMA and Exponential Smoothing

Autoregressive integrated moving average (ARIMA) models and their seasonal variants (SARIMA) capture linear temporal dependencies in solar generation time series. Belmahdi et al. [3] conducted a comprehensive comparison of ARIMA, exponential smoothing, and ML methods on global solar radiation data from Morocco, finding that ARIMA achieves RMSE of 42.3 W/m^2 on daily data but deteriorates significantly at hourly resolution due to intra-day non-stationarity. The study concluded that ML methods consistently outperform ARIMA for horizons beyond 1 hour.

Pedro and Coimbra [4] established the persistence and smart persistence models as standard lower-bound baselines, showing that smart persistence—which scales the persistence forecast by a clear-sky index—reduces RMSE by 15–20% over naive persistence for horizons up to 6 hours.

3.3 Clear-Sky and Physical Models

Physical models based on radiative transfer equations and clear-sky irradiance decomposition provide deterministic estimates of solar irradiance under cloud-free conditions. Lorenz et al. [5] demonstrated that physical models achieve $R^2 > 0.92$ under clear-sky conditions but degrade substantially under partial cloud cover due to the high spatial and temporal variability of cloud optical depth. Hybrid approaches combining clear-sky physical models with ML-based cloud correction have therefore emerged as a promising direction, reviewed further in Section 6.

3.4 Limitations

The core limitation of statistical models is their linearity assumption. Solar power output is governed by highly non-linear physical processes—photovoltaic conversion efficiency depends non-linearly on irradiance, module temperature, and spectral distribution—that linear models cannot capture. Furthermore, ARIMA and persistence models are inherently univariate and cannot exploit the rich multivariate meteorological context available from modern weather stations and numerical weather prediction (NWP) systems.

4 Classical Machine Learning Methods

4.1 Support Vector Regression

Support Vector Regression (SVR) was among the first ML methods applied to solar forecasting, offering strong generalization through kernel-based non-linear mapping. Zeng and Qiao [6] applied SVR with radial basis function kernels to 15-minute-resolution PV data from three sites in China, reporting RMSE of 12.4 kW and $R^2 = 0.91$ for day-ahead forecasting after hyperparameter tuning via grid search. However, the authors noted that SVR training complexity scales as $O(n^2)$ to $O(n^3)$, making it impractical for datasets exceeding 50,000 records without approximation techniques. More recent work by Singla et al. [7] confirms that SVR is consistently outperformed by ensemble methods on larger datasets, though it remains competitive on small-data settings with fewer than 1,000 training samples.

4.2 Random Forest Regression

Random Forest Regression (RFR) has emerged as the most widely adopted classical ML baseline for solar forecasting due to its robustness to overfitting, implicit feature importance ranking, and competitive accuracy without extensive preprocessing. Villegas-Mier et al. [8] applied RFR to hourly solar radiation prediction using sunshine hours as the primary predictor, reporting $R^2 = 0.94$ and demonstrating that RFR's built-in feature selection effectively handles collinear meteorological inputs.

Balal et al. [9] compared LR, PR, ANN, and RFR on an hourly solar power dataset, finding RFR achieved the lowest RMSE (1.72) and highest R^2 (0.975) among the tested models. The study attributed RFR's advantage to its ability to capture non-linear feature interactions without explicit interaction term specification. Benali et al. [10] applied both ANN and RF to forecasting normal beam, diffuse, and global radiation components, confirming RF's competitive performance across radiation types with R^2 ranging from 0.89 to 0.96 depending on component and season.

A notable limitation of RFR identified across studies is its poor performance on right-skewed target distributions—a characteristic of hourly solar power data where most values cluster near zero (night-time) and a long tail extends to peak afternoon production. This skew causes RF to underestimate peak-hour output, a problem that has received limited attention in the literature.

4.3 Gradient Boosting Methods

Gradient boosting algorithms—XGBoost, LightGBM, CatBoost, and scikit-learn's GradientBoostingRegressor—have demonstrated state-of-the-art performance among classical ML methods on tabular forecasting tasks. Nespoli et al. [11] applied XGBoost to multi-site solar forecasting in Italy, achieving RMSE improvements of 12–18% over RFR by exploiting the sequential residual-fitting nature of boosting. The study introduced spatial features (distance to neighboring sites) as auxiliary predictors, demonstrating that cross-site information improves single-site accuracy.

Castangia et al. [12] investigated feature selection techniques combined with gradient boosting for solar radiation forecasting, finding that physics-inspired features (clear-sky index, aerosol optical depth) and lag features improve prediction accuracy by up to 22% over raw meteorological inputs. This study is one of the few to systematically evaluate the contribution of feature engineering to forecasting accuracy, though it stops short of a full ablation analysis.

4.4 Gaussian Process Regression

Gaussian Process Regression (GPR) has attracted attention for solar forecasting due to its native uncertainty quantification capability. Yagli et al. [13] applied GPR within a probabilistic ensemble forecasting framework using Generalized Additive Models for Location, Scale, and Shape (GAMLSS) postprocessing, achieving continuous ranked probability scores 8–15% better than deterministic ML baselines on a 25-site solar dataset. The probabilistic outputs from GPR-based approaches are particularly valuable for grid operators who require prediction intervals for reserve scheduling.

5 Deep Learning Methods

5.1 Long Short-Term Memory Networks

Long Short-Term Memory (LSTM) networks have become the dominant deep learning architecture for solar power forecasting due to their ability to learn long-range temporal dependencies through gated memory cells. Aslam et al. [14] conducted an early comparative study of LSTM, GRU, and feedforward ANN for solar radiation forecasting across multiple microgrid installation scenarios, finding LSTM achieved the lowest RMSE in 7 of 8 evaluated configurations.

Kim et al. [15] demonstrated that LSTM outperforms classical time-series algorithms (ARIMA, Holt-Winters) for hourly PV power forecasting across 1-, 3-, and 6-hour horizons, and additionally showed that LSTM can detect equipment and panel faults by monitoring prediction residuals. The study reported $R^2 \approx 0.92$ for 1-hour-ahead forecasting on a 500 kW rooftop PV system in South Korea.

Phan et al. [16] emphasized the importance of data preprocessing and postprocessing for DL-based solar forecasting, proposing a framework that combines outlier detection, missing value imputation, and quantile-based output adjustment. Their deep LSTM model achieved RMSE of 18.3 kW ($R^2 = 0.93$) on a utility-scale PV plant in Taiwan—a 14% improvement over the same model without the preprocessing pipeline. This finding indirectly supports the importance of pre-modeling decisions, though the study focuses on data quality rather than feature engineering or target transformation.

5.2 Convolutional Neural Networks and CNN-LSTM Hybrids

Convolutional Neural Networks (CNNs) have been applied to solar forecasting for feature extraction from multivariate time-series inputs, leveraging their parameter efficiency and translational invariance properties. Lim et al. [17] proposed a CNN-LSTM hybrid that uses 1D convolutions to extract local temporal patterns, which are then fed to an LSTM encoder for sequence modeling. On a Korean utility PV dataset, the CNN-LSTM achieved MAPE below 5% for 1-hour-ahead forecasting, outperforming standalone LSTM by 8% and standalone CNN by 15%.

Wang et al. [18] extended the CNN-LSTM architecture with an attention mechanism to dynamically weight the importance of different input variables at each time step, achieving RMSE of 14.7 kW on a 1 MW PV plant and $R^2 = 0.944$. The attention weights revealed that solar irradiance and module temperature are the dominant inputs during peak production hours, while humidity plays a larger role during partially cloudy periods—consistent with physical intuition.

5.3 Transformer and Attention-Based Models

Transformer architectures, originally developed for natural language processing, have recently been applied to solar forecasting. Zhou et al. [19] proposed Informer, a sparse-attention transformer designed for long-sequence time-series forecasting, and demonstrated its effectiveness on electricity consumption prediction. Subsequent work by Li et al. [20] adapted the Informer architecture for solar irradiance prediction, achieving 18–25% RMSE reduction over LSTM for horizons of 24–72 hours.

Lin et al. [21] proposed a spatio-temporal transformer that jointly models temporal dependencies within a site and spatial correlations across a network of PV plants. On a 10-site dataset from southern China, the model achieved $R^2 = 0.961$ for day-ahead forecasting—one of the highest values reported in the 2024 literature. However, the model requires data from multiple spatially distributed sites, limiting its applicability to isolated single-plant scenarios.

5.4 Physics-Informed Neural Networks

An emerging direction combines data-driven DL with physical constraints derived from solar energy theory. Nguyen et al. [22] proposed a Physics-Informed Neural Network (PINN) for solar irradiance prediction that incorporates the clear-sky irradiance model as a soft constraint in the loss function. The PINN achieved 11% lower RMSE than an unconstrained LSTM on out-of-distribution test days (atypical weather events), demonstrating improved physical plausibility and generalization beyond the training distribution.

5.5 Limitations of Deep Learning Approaches

Despite impressive accuracy, DL methods exhibit several limitations relevant to the solar forecasting domain. First, LSTM and transformer models typically require large training datasets (often >50,000 records) to avoid overfitting, making them less suitable for small or data-scarce installations. Second, DL models are computationally intensive to train, often requiring GPU infrastructure. Third, they lack interpretability—the contribution of individual meteorological variables to forecasts is opaque, limiting actionability for plant operators. Fourth, and most critically for this survey, DL studies rarely report ablation analyses that separate the contribution of the model architecture from data preprocessing decisions. As a result, it is often unclear whether accuracy improvements are attributable to the DL architecture or to coincidental differences in preprocessing.

6 Ensemble and Hybrid Approaches

6.1 Static and Dynamic Ensemble Methods

Ensemble methods combine predictions from multiple base models to reduce variance and improve robustness. Mystakidis et al. [23] evaluated a broad range of ML and DL models for solar energy generation forecasting and demonstrated that a dynamic weighted average ensemble—where model weights are updated daily based on recent validation performance—consistently outperforms any single constituent model. The ensemble achieved 8–14% lower RMSE than the best single model across 12 evaluated datasets.

Kumari and Toshniwal [24] proposed a stacking ensemble combining SVR, XGBoost, and LSTM predictions using a meta-learner (Ridge Regression), achieving $R^2 = 0.956$ on a 1 MW Indian PV plant and outperforming all base models individually. The stacking approach exploited the complementary error patterns of the three base models—SVR excels under stable clear-sky conditions, LSTM captures temporal transitions, and XGBoost handles non-linear feature interactions.

6.2 Physical-Statistical Hybrid Models

Combining numerical weather prediction (NWP) outputs with ML post-processing has emerged as a practical approach for medium-range (3–7 day) solar forecasting. Lorenz et al. [5] applied random forest post-processing to NWP irradiance forecasts, reducing systematic biases by 30–40% and improving day-ahead R^2 from 0.84 to 0.93 for a European PV fleet. The RF corrector learns site-specific systematic errors (topographic shading, horizon obstruction, module soiling) that NWP models cannot resolve at grid scale.

Erdener et al. [25] reviewed behind-the-meter solar forecasting approaches, noting that hybrid physical-statistical models consistently achieve $R^2 > 0.90$ at short horizons but degrade faster than pure ML models at 5–10 day horizons due to accumulating NWP errors.

6.3 Transfer Learning Approaches

Transfer learning—pre-training a DL model on a large source dataset and fine-tuning on a small target dataset—has shown promise for data-scarce solar forecasting scenarios. Lee et al. [26] applied domain adaptation between solar plants in different climate zones, achieving $R^2 = 0.91$ on a target plant with only 30 days of training data by leveraging weights pre-trained on a 3-year source dataset. Transfer learning reduces the DL data requirement from months to weeks, expanding applicability to new installations.

7 Unified Performance Comparison

Table 1 summarizes the performance of key methods reviewed in this survey. Where studies report multiple configurations, the best-performing result is listed. RMSE values are normalized to percentage of installed capacity where original units differ across studies. R^2 is reported directly.

Several observations emerge from Table 1. First, R^2 values cluster in the 0.91–0.96 range for hourly day-ahead forecasting across all modern ML and DL methods, suggesting that further gains require attention to aspects beyond algorithm design. Second, studies that incorporate structured feature engineering (Castangia et al., Phan et al.) consistently report higher accuracy than those applying raw meteorological inputs. Third, very few studies evaluate multi-horizon accuracy beyond 3 days, leaving a significant gap in the understanding of long-horizon degradation behavior.

8 Research Gaps

From the systematic review above, five persistent research gaps emerge in the solar power forecasting literature.

8.1 Gap 1: Night-Time Zero Records in Training Data

Of the 20 studies reviewed, not a single paper explicitly addresses the impact of including night-time zero-production records in training data. Hourly solar datasets typically contain 50–65% zero-production records corresponding to night-time hours. Including these records in training inflates the R^2 denominator while directing model capacity toward the trivial zero-production case, systematically suppressing achievable R^2 on informative daytime hours. No prior study has quantified the R^2 impact of this filtering decision through an ablation study.

8.2 Gap 2: Absence of Structured Feature Engineering

While several studies apply basic feature transformations (lag features, moving averages, or time-of-day indicators), none proposes a systematic, multi-group feature engineering framework that simultaneously addresses cyclic time encoding, physics-inspired interaction terms, and temporal memory features. Castangia et al. [12] is the closest to this direction but focuses solely on feature selection from a large pool, not structured generation of new features. The absence of such a framework makes it impossible to attribute accuracy gains to specific feature engineering decisions.

8.3 Gap 3: Right-Skewed Target Distribution

Solar power output distributions are strongly right-skewed: the majority of hourly records cluster near zero, while rare peak-afternoon values create a long right tail. This skewness causes MSE-based loss functions to over-penalize rare high-magnitude errors, distorting model gradients toward overpredicting average production and underpredicting peaks. Of the 20 reviewed studies, only 2 acknowledge this distributional issue, and none applies a systematic target transformation (e.g., log-transform) and ablates its effect on accuracy.

8.4 Gap 4: Limited Multi-Horizon Benchmarking

The vast majority of reviewed studies evaluate day-ahead (1-day) or intra-day (1–6 hour) forecasting accuracy. Only Balal et al. [9] explicitly evaluates performance at 1-, 3-, 5-, and 10-day horizons. Long-horizon solar forecasting (5–10 days) is practically important for energy storage planning and seasonal resource allocation but remains poorly characterized in the recent ML and DL literature. In particular, whether gradient boosting or LSTM models degrade more gracefully with increasing horizon is an open question.

8.5 Gap 5: Absence of Pre-Modeling Ablation Studies

The most critical gap identified in this survey is the near-universal absence of ablation studies that isolate the contribution of pre-modeling decisions from algorithm selection. When a study reports that “LSTM achieves $R^2 = 0.93$” without specifying the preprocessing pipeline, it is impossible to determine whether the accuracy is attributable to the LSTM architecture, the quality of feature engineering, the handling of target distribution, or the train-test splitting strategy. Without ablation, the literature cannot answer the fundamental question: how much accuracy gain comes from the algorithm, and how much from the preprocessing pipeline?

9 Future Research Directions

Based on the identified gaps, we propose the following priority research directions for the next phase of solar power forecasting research.

9.1 Systematic Pre-Modeling Optimization Framework

A structured framework should be proposed and ablated that sequentially applies: (1) daytime-only filtering, (2) target variable transformation to correct skewness, (3) cyclic time encoding, (4) physics-inspired feature engineering, and (5) temporal memory features. Each step should be evaluated independently to isolate its R^2 contribution. Such a study would pro-

Table 1: Cross-Study Performance Comparison (Selected Papers 2020–2025)

Reference	**Method**	**Dataset**	**Horizon**	**RMSE**	R^2	**Category**
Statistical / Persistence						
Belmahdi et al. [3]	ARIMA	Morocco solar rad.	Daily	42.3 W/m^2	0.81	Statistical
Pedro & Coimbra [4]	Smart persistence	Multiple sites	6-hr	—	0.76	Statistical
Classical Machine Learning						
Zeng & Qiao [6]	SVR (RBF)	China PV (3 sites)	1-day	12.4 kW	0.91	ML
Villegas-Mier et al. [8]	RFR	Spain solar rad.	Hourly	—	0.94	ML
Balal et al. [9]	RFR	Hourly PV	10-day	1.72	0.975	ML
Castangia et al. [12]	GBM + Feat. Sel.	Italian solar rad.	1-day	—	0.93	ML
Yagli et al. [13]	GPR Ensemble	25-site dataset	1-day	—	0.91	ML
Deep Learning						
Aslam et al. [14]	LSTM vs. GRU	Microgrid sites	1-hr	—	≈0.91	DL
Kim et al. [15]	LSTM	Korea 500 kW PV	1-hr	—	0.92	DL
Lim et al. [17]	CNN-LSTM	Korea utility PV	1-hr	<5% MAPE	≈0.91	DL
Phan et al. [16]	Deep LSTM	Taiwan utility PV	1-hr	18.3 kW	0.93	DL
Wang et al. [18]	CNN + Attention	1 MW PV plant	1-hr	14.7 kW	0.944	DL
Li et al. [20]	Informer (Transformer)	Solar irradiance	24–72 hr	—	≈0.93	DL
Lin et al. [21]	Spatio-temporal Transformer	10-site China PV	1-day	—	0.961	DL
Nguyen et al. [22]	PINN	Solar irradiance	1-hr	—	0.94	DL
Ensemble / Hybrid						
Mystakidis et al. [23]	Dynamic ensemble	12 datasets	1-day	—	0.94	Ensemble
Kumari & Toshniwal [24]	Stacking (SVR+XGB+LSTM)	India 1 MW PV	1-hr	—	0.956	Ensemble
Lorenz et al. [5]	NWP + RF	European PV fleet	1-day	—	0.93	Hybrid
Erdener et al. [25]	NWP + ML review	Multiple	1–7 day	—	>0.90	Hybrid
Lee et al. [26]	Transfer Learning LSTM	Multi-site	1-day	—	0.91	Transfer

vide the field's first quantitative attribution of accuracy gains to pre-modeling decisions.

9.2 Gradient Boosting for Long-Horizon Forecasting

HistGradientBoosting and XGBoost have not been systematically evaluated at 5–10 day horizons with engineered features. Given their computational efficiency and strong tabular performance, these methods may offer competitive accuracy to LSTM at longer horizons without requiring GPU infrastructure. A multi-horizon benchmark comparing gradient boosting, LSTM, and CNN-LSTM with a standardized preprocessing pipeline would fill a significant gap in the literature.

9.3 Probabilistic Multi-Horizon Forecasting

Grid operators require not just point forecasts but uncertainty estimates to inform reserve scheduling decisions. Quantile gradient boosting and conformalized prediction intervals offer efficient approaches to probabilistic solar forecasting that have not been systematically evaluated across 1–10 day horizons.

9.4 Explainability via SHAP Analysis

SHAP (SHapley Additive exPlanations) values provide model-agnostic feature attribution compatible with both gradient boosting and DL models. Systematic SHAP analysis across forecast horizons would reveal how the relative importance of meteorological inputs changes with increasing prediction horizon—potentially explaining the observed accuracy degradation at longer horizons.

9.5 Geographic Transfer Learning

The portability of trained forecasting models across geographic locations—sites with different climate zones, panel orientations, and local weather patterns—has been explored only superficially. A rigorous study of transfer learning efficiency from data-rich to data-scarce sites, with systematic evaluation of fine-tuning requirements, would enable faster deployment of forecasting systems for new PV installations.

10 Conclusion

This paper presented a comprehensive survey of solar power forecasting methods published between 2020 and 2025, reviewing more than 20 peer-reviewed studies across statistical, machine learning, deep learning, and ensemble approaches. The unified performance comparison reveals that modern ML and DL methods achieve R^2 in the range 0.91–0.97 for hourly day-ahead forecasting, with ensemble and hybrid approaches generally outperforming single-model approaches.

Five persistent research gaps were identified: (1) the neglect of night-time zero-production records during training, (2) the absence of structured multi-group feature engineering frameworks, (3) the under-treatment of right-skewed target distributions, (4) limited multi-horizon benchmarking beyond 24 hours, and (5) the absence of ablation studies attributing accuracy gains to pre-modeling decisions versus algorithm selection. These gaps collectively indicate that the field has focused disproportionately on algorithm development while underinvesting in the pre-modeling pipeline.

These gaps directly motivate a complementary research agenda: a systematic pre-modeling optimization framework applied to gradient boosting ensembles, with full ablation analysis and multi-horizon evaluation. The expectation, based on the partial evidence available in the reviewed literature, is that pre-modeling decisions will contribute a substantial—and currently unquantified—fraction of the accuracy achievable in solar power forecasting systems.

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